THERMODYNAMICS & HVAC SYSTEMS GUIDED INQUIRY ACTIVITY

Thermodynamic Analysis of Air-Water Systems

A Guided Inquiry Exploration of the Psychrometric Chart

Name: _____ Date: _____

LEARNING OBJECTIVES

- Identify, read, and define the seven key thermodynamic properties represented on a psychrometric chart.
- Determine the state point of moist air given any two independent properties.
- Trace and analyze standard thermodynamic processes (heating, cooling, dehumidification, and adiabatic mixing).
- Evaluate the impact of atmospheric pressure (elevation) on psychrometric properties.
- Distinguish between saturated enthalpy (h_{sat}) and actual enthalpy using enthalpy correction curves (Δh).

1. Introduction: The Geometry of Air

A psychrometric chart is a graphical representation of the thermodynamic state of a mixture of dry air and water vapor. Under Gibbs' Phase Rule, a mixture of two gases can be completely specified by three independent intensive properties. If we fix the total pressure (P), we can plot any state point on a two-dimensional coordinate space using any two other independent variables.

Task 1 Orienting Yourself on the Grid

Open the **Psychrometric Chart Simulator**. Set the elevation to **0 m** (Sea Level) and toggle the units to **Metric (SI)**. Locate the primary axes and boundary curves of the interactive plot, and answer the following:

a. What thermodynamic property is mapped to the horizontal axis (Abscissa)? State its variable symbol and metric unit.

b. What property is mapped to the vertical axis on the right side of the plot (Ordinate)? State its variable symbol and unit. Why is it scaled in g/kg of dry air instead of simply kg/kg?

c. Hover your cursor along the far-left boundary curve where Relative Humidity is exactly 100%. What is this

curve called, and what physical transition does it represent?

2. Essential Elements of the Chart

By using the interactive legend highlighting buttons at the bottom of the simulator, systematically hover over each curve type and analyze its trajectory.

Task 2 Tracing and Defining State Properties

Fill out the table below to catalog the seven primary moist air state variables, their geometric representations, and their behaviors on the chart.

Thermodynamic Property	Symbol	Metric Unit	Orientation of Constant-Value Lines
Dry Bulb Temperature	T_{db}	°C	*e.g., Straight vertical lines...*
Humidity Ratio	W	g/kg_{da}	
Relative Humidity	ϕ (or RH)	%	
Wet Bulb Temperature	T_{wb}	°C	
Dew Point Temperature	T_{dp}	°C	
Specific Enthalpy	h	kJ/kg_{da}	
Specific Volume	v	m^3/kg_{da}	

Inquiry Questions: Physical Intuition

1. Why do the constant specific volume lines slope downwards to the right? As Dry Bulb Temperature increases at a constant Humidity Ratio, what happens to specific volume, and how does this match your understanding of the Ideal Gas Law?

2. Observe the wet-bulb temperature lines (T_{wb}) and the enthalpy lines (h). They are nearly, but not *exactly*, parallel.

i. Turn on the **Enthalpy Correction (Δh Deviation)** curves using the legend. Describe what these curves represent.

ii. Choose a state point of T_{db} = 35°C and RH = 20%. Read the Saturated Enthalpy on the diagonal scale, and use the hover reticle to find the actual enthalpy and the deviation (Δh). Why is there a difference?

3. Locating Moist Air States

Rule of Two Properties: To find any state point on the chart, you must know the atmospheric pressure (P) and **at least two** other independent intensive thermodynamic properties.

Task 3 Determining States at Sea Level

Set the elevation to **0 m** (P = 101.3 kPa). Use the simulator's input panel or click-to-plot capability to determine the full thermodynamic properties of the following air states:

- **State 1 (Warm & Moist room air):** T_{db} = 26.0°C, φ = 60%
- **State 2 (Cooler coil output air):** T_{db} = 12.0°C, T_{wb} = 10.0°C

State	T_{db} (°C)	φ (%)	W (g/kg)	T_{wb} (°C)	T_{dp} (°C)	h (kJ/kg)	v (m^3/kg)
State 1	26.0	60.0	_____	_____	_____	_____	_____
State 2	12.0	_____	_____	10.0	_____	_____	_____

Task 4 The Impact of Elevation & Pressure

Now, change the elevation in the top bar from **0 m** to **2,000 m** (representative of a high-altitude city like Mexico City).

1. What is the new atmospheric pressure reported by the simulator?

P = _____ kPa

2. Keep the temperature of State 1 fixed at T_{db} = 26.0°C and Humidity Ratio fixed at W = 10.0 g/kg. Switch the simulator calculation type to "Dry Bulb + Humidity Ratio" and enter those parameters. Report the new properties:

- `Relative Humidity (ϕ):` _____ %
- `Enthalpy (h):` _____ kJ/kg
- `Dew Point Temp (T_dp):` _____ °C
- `Specific Volume (v):` _____ m³/kg

3. Explain physically why Specific Volume (v) and Relative Humidity (ϕ) shifted when you reduced the total air pressure, even though temperature and absolute moisture mass fraction remained identical.

_____

4. HVAC Process Analysis

By selecting different modes under the **Process Simulator** menu, trace the path lines connecting State 1 and State 2 and apply conservation laws to evaluate air treatments.

Task 5 Sensible Heating and Cooling

Set elevation to **0 m**. Switch the process simulator to **Sensible Heating / Cooling**. Configure State 1 to T_{db} = 15.0°C, φ = 50%. Configure State 2 to T_{db} = 35.0°C.

1. Describe the geometry of the process path line on the chart. What is the slope?

2. Explain why the Humidity Ratio (W) and Dew Point Temp (T_{dp}) must remain perfectly constant during a purely sensible thermal process.

3. Relative humidity changed from _____% down to _____%. Why did the air become "drier" when no moisture was removed?

Task 6 Adiabatic Humidification (Evaporative Cooling)

Switch the simulator to **Adiabatic Humidification**. Configure State 1 to a hot, dry climate state of T_{db} = 38.0°C, φ = 15%. Run the evaporative cooling process down to a comfort level of T_{db} = 24.0°C.

1. Trace the process direction. What curve or property line does this process closely follow? Why?

2. Find the resulting Humidity Ratio at State 2: W_2 = _____ g/kg. Calculate the total amount of water vapor added to each kilogram of dry air:

$\Delta W = W_2 - W_1 =$ _____ g of water / kg of dry air

3. What is the absolute lowest temperature (limit of cooling performance) achievable by an evaporative cooler operating on this incoming air? What thermodynamic name is given to this limit?

Task 7 Stream Mixing (Mass & Energy Balancing)

Switch the simulator to **Adiabatic Mixing of Air Streams**. Set State 1 (Return Air) to T_{db} = 25.0°C, φ = 50%. Set State 2 (Outside Air) to T_{db} = 5.0°C, φ = 80%.

1. Observe the resulting State 3 (Mixture) on the graph. Explain geometrically where State 3 resides relative to the points of State 1 and State 2.

2. Adjust the **Stream 1 Mass Ratio** slider to 60% (Return Air) and 40% (Outside Air). Report the calculated properties of the mixed air entering our handler:

- $T_{db,3}$: _____ °C
- W_3: _____ g/kg
- h_3: _____ kJ/kg

3. Prove mathematically using mass and energy balances that: $T_{db,3} = r_1T_{db,1} + r_2T_{db,2}$ and $h_3 = r_1h_1 + r_2h_2$. Show your equations and calculations. Do they match the simulator's output?

5. Synthesis Challenge: Designing Comfort

Imagine you are designing the cooling system for an office building in Denver, Colorado (elevation ≈ 1,600 m). The indoor space must be maintained at T_{db} = 22.0°C and φ = 50%. The outdoor air is at T_{db} = 34.0°C and T_{wb} = 14.0°C (hot, dry alpine climate).

1. Determine the outdoor relative humidity (φ_{out}) and specific volume (v_{out}) at Denver's altitude.

- ϕ_{out}: _____ %
- v_{out}: _____ m³/kg

2. Based on the low outdoor wet-bulb temperature, can an indirect evaporative cooler alone handle the cooling load to reach 22°C indoors without mechanical refrigeration? Defend your design decision using chart properties.